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Detection of Disease in Tea Leaves Using Convolution Neural Network

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***Abstract*—Tea leaf diseases such as *Black Rot* and *Rust of Tea* are a major critical threat to Indian food security. The reason being is rapid identification of these diseases still remains difficult in many corners of the world till today because it requires huge amount of work, prior knowledge in the concerned diseases, and also requires the huge processing time. Diseases are always injurious to the tea plant's health which in turn severely affect its growth. In 2018, tea industry contributes almost 6% in the Indian economy. So it is always advisable to continuously monitor its growth to ensure minimal losses of this tea plant. But it is not possible to monitor this using naked eye. So this paper basically adopts convolutional neural network (CNN) model to detect and identify these two disease using convolutional neural network (CNN). The proposed works basically finds a solution of the tea leaf disease detection using simplest method while keeping minimum computational complexity and minimal resource to gain fast and accurate result as convolutional neural network (CNN) automatically extracts features for classification of input image into various classes. The experimental results on the developed model achieved precision approximately 95.93%.**

***Index Terms*—Tea leaf disease detection, Convolution Neural Network, Deep Learning, Machine Learning, Image Processing.**

I. Introduction

Technology plays an important role for the human beings when it comes to the production of food in a country like India whose economy depends largely on agriculture. Tea is one the most consumed beverage in India as it contain more than 700 different chemicals like flavanoides, amino acids, vitamins C, Vitamin E, Vitamin K, caffeine and polysaccharides which are extremely health beneficial. Most Surprisingly the vitamin C content in tea leaves is almost comparable to the citrus fruits. So if the leaf is infected it may be very risky to consume it as our health totally depends on the food we consume. So healthy plants is an important step of good crop productivity ([14], [15]). One of the best old way to overcome this is by using pesticide like fertilizers or its equivalent to combat the effect of those specific pathogens if diseases are correctly identified and diagnosed early for fast growth. But this kind of fertilizers also affect our body if left unattended on time. Hence the biggest challenge for tea cultivators nowadays worldwide is to produce tea without use of pesticide residues and this growing demand for residue free tea in the highly industrialised countries can be fulfilled using technology.

However manual monitoring cannot be so effective on a single glance for prediction purposes and also it is quite time consuming. But nowadays it is possible to overcome some problems using advance technological machine support. In Machine learning (ML), a prerequisite data is imposed to a system and make that system to learn by its own mechanism and then apply its learned model to perform a given task. Deep learning (DL) is a sub part of Machine learning (ML). Deep learning (DL) has gained its importance due to his high end computational power. Different methods can be applied for learning the task such as supervised learning, unsupervised learning and Reinforcement learning. In the supervised learning method training of model is done using labelled data. In supervised, no supervision is needed for training. Supervised learning means collect data and produced output depending on past experience.

II. Related Work

It is very important to study the already existed research present in this field to move forward correctly in the right path. Tea leaf disease detection has been a critical research domain in which image processing techniques and DL techniques both have been used for its accurate detection and classification.

The aim of this work [2] is to identify and classify Banana leaf diseases, like Banana speckle and Banana sigatok by performing adequate training of samples using DL models under certain critical conditions like complex dark background, different images resolution, different illumination, different size and orientation angle. This paper basically demonstrate the accuracy of this method with very less computational efforts.

In this paper [3], Hughes et al. propose detection of diseases using DL techniques and for detection purposes various dataset containing a collection of numeorus number of images of different types of crops, along with their affected and unaffected images. Here two DL models are used namely GoogleLeNet and AlexNet, which provides almost an accuracy of 99.3%. But this model suffers with very less classification rate when backgroud contrasting of images is not done properly.

The main aim of paper [4] is to classify various tomato leaves diseases into number of classes like Bacterial canker, Bacterial spot etc. A sample of almost 383 images have been captured using a digital camera. Then Otsu's method has been used on the collected samples for image segmentation purposes. Color features, texture features, shape features have been obtained. Later all these features have been combined to form a final feature module. Finally Supervised learning methods have been applied for classification of diseases by training the classifier. Here the classifier used is decision tree. Although it gives very high accuracy but this model suffers from over fitting when the data is noisy.

In paper [5], Sladojevic et al. propose a model to classify a count of 13 various types of crop diseases, but the samples they used in this paper consist of resized, higher resolution, good illuminated and cropped images. To overcome the over-fitting issue, process of augmentation was also incorporated and finally provides an accuracy of 96.3%

Here, in [6], Mokhtar et al. propose a method using Gabor wavelet transformation (GWT) technique to extract various features in the disease detection and identification of tomato leaves. This extracted features were then given to some Support Vector Machine (SVM) classifier for training purposes of the model. All the noise cancellation, cropping, resizing and background clearance are done in the pre processing step. Gabor transformation is used to detect the textual patterns of the infected leaves and extract adequate important features required for identification and then the classification was done using Support Vector Machine (SVM) mechanism with various kernel functions. This model provides an accuracy of about 99.5% as showed. But this model has a limitation due to the use of Gabor Transformation as it is very much computationally intensive.

In [7], Dyrmann et al. describe a model of only coloured images and Deeper Network architecture model is implemented to detect weed, a total of 22 different types of weeds have been classified with an accuracy of about 86.2%. But this model suffers drawback due to less accuracy.

In paper [8], early stage symptoms detection is done using analytical result derived from its color specification. Along with that, K-means clustering algorithm has also been implemented for the identification of the diseased pixel and lastly, Fuzzy Logic has been implemented for disease classification.

In paper [10], Karmokar et al. have proposed a tea leaf disease recognizer that combines feature extraction and neural network ensemble for training. First step is the acquisition of tea leaf images by digital camera. Next step is the pre-processing of the dataset. In this step, images are converted into threshold images. Next step is the feature extraction to get matrix representation of the extracted image. Training of these dataset is done by negative correlation algorithm. The negative correlation algorithm is one of the methods to combine the predictions of the various independent trained neural networks. Next step is the testing that is recognizing the tea leaf disease.

III. PROPOSED METOHODOLOGY

The model is designed to detect the unhealthy leaves of disease like Black Rot and Rust of tea. There are two phase of this model: one phase is training phase and the other is testing phase. In training phase, first clear image of the tea leaf is collected, then image labelling and pre processing and lastly CNN based training was done. In the next phase, again new testing image collected, then image preprocessing and lastly disease identification and classification. A total of 2341 images are collected including healthy and unhealthy leaves of Black rot and Rust of tea disease. A total of 3 classes have been created. The following figures portrays the overall idea of the model.

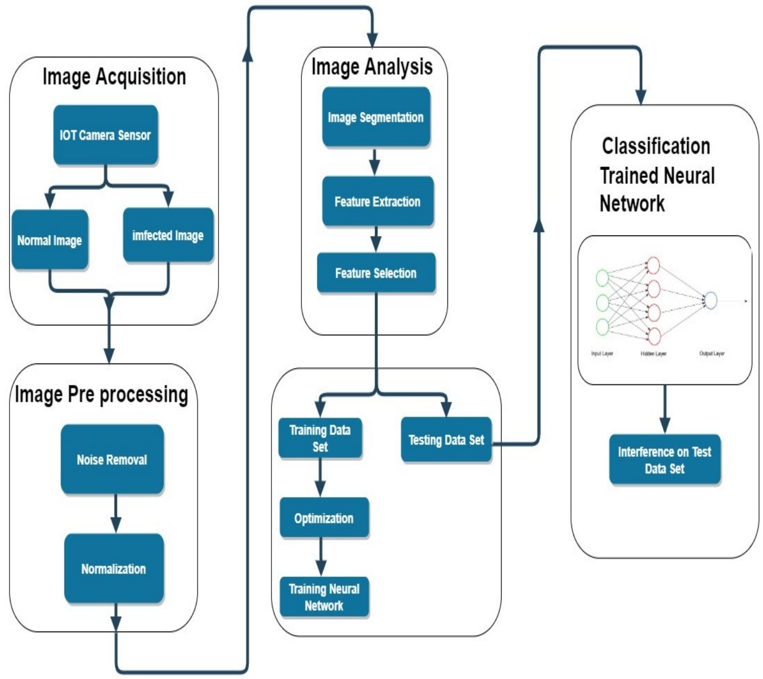


Fig. 1: Block diagram of the proposed model

The entire procedure of designing the model for tea plant disease recognition using CNN is stated in the above block diagram and it is divided into several steps starting with collection of images for classification process using deep neural networks.

A. Dataset

The already existing databases such as ImageNet, PlantVillage[1], and CIFAR-1 datasets management do not have adequate tea leaf disease images and some studies have collected disease photos in various controlled and indoor environments. Hence due to these factors the disease recognition system under natural light conditions have certain limitations, so a new disease data set has been constructed in this paper.

Majority of the Tea leaf disease images were all captured using the Realme AI powered 64Mp Quad Camera in the natural light environment of the local tea garden in Local Tea Estate of Assam and some are collected from TTRI ([12], [13]) . The images were resized about 20 cm directly with autofocus mode at resolution of 512 × 512 pixels. A total of 2341 images were collected, which contained 3 diseases, and all disease images

TABLE I: Dataset Samples

Class	*Total No of Images*	*Images taken for training out of total images*	*Images taken for testing out of total images*
Healthy Tea Leaf	1245	500	100
Black Rot of Tea	621	500	100
Rust of Tea	475	300	100
Total	2341	1300	

have been identified by plant pathologists.
Table 1 shows the number of images for every class used as training, validation and testing datasets for the disease classification model. The disease classification datasets that were used in these analyses are shown in Table 1. The 80/20 ratio of training/test data is the most commonly used ratio in neural network applications. In addition, a 10% subset of the test dataset was used to validate the dataset

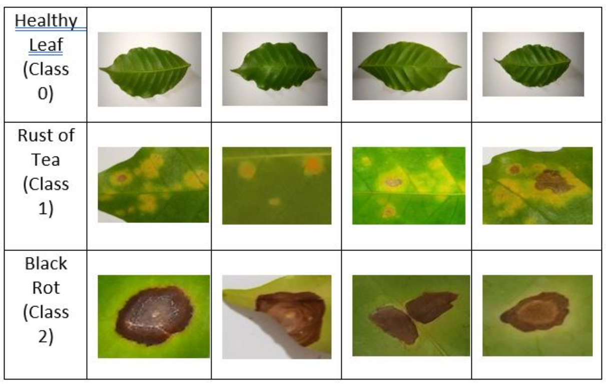


Fig. 2: Some samples of tea leaves from datasets.

B. Image Preprocessing and labelling

- Images captured from field were in different quality and resolutions.
- In order to get better feature extraction, all the collected images were preprocessed for a better consistency.
- However it means augmenting and cropping of all the collected images manually, making a square around all the leaves (infected and healthy), in order to specifically highlight the region of interest (ROI) of the images.
- Image samples with lesser resolution and dimension not more than 500 pixels were not considered as valid images for traning purposes. Only those image samples whose region of interest (ROI) of the images was in higher resolution and pixels more than 500 were considered as eligible samples for training purposes
- All the image samples used for the dataset were again resized to $256 * 256$ in order to reduce the time required for training.

C. Augmentation

- Here slight variation and distortion to the image samples is applied to overcome the overfitting problem during the training stage.
- Some of the augmented techniques are:

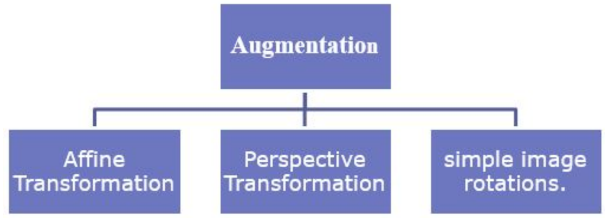


Fig. 3: Augmented Techniques

- Here in order to automate the augmentation process for numerous images from the dataset, there is a process of changing the parameters of transformation during the run-time, which improves flexibility.

D. CNN Training

- Training the Convolutional Neural Network (CNN) for making an image classification model from a dataset was proposed.
- Caffe Net[9] is basically a deep CNN which has multiple type of layers that progressively extract features from the input samples.

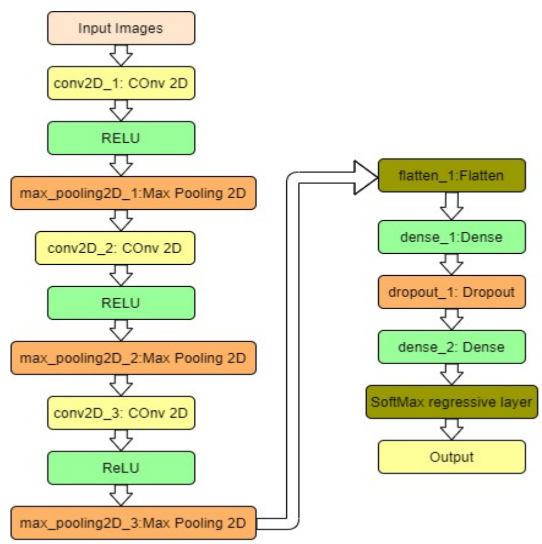


Fig. 4: Model architecture

- Deep Convoliution Neural Network (CNN) with Rectified Linear Units (ReLUs) trains several times faster. This method is applied to the cascaded output of every convolutional and fully connected layer. The model used in this paper has been shown in Fig no. 4 [11].
- Then comes another important layer of Convoliution Neural Network (CNN) which is the pooling layer and it

is basically a configuration of non linear type down sampling. Pooling operation gives the form of translational invariance.

IV. Experimental Analyses

The collection of data for our working model is basically added from some real-time scenario data from field exhibiting different properties. Hence it is very limited. Some of the data samples are shown in Fig. 2 and Fig. 5. So in order to radically expand the dataset samples, augmentation of data has been implemented randomly that means rotating the image samples by a marginal amount of 10 degrees or 20 degree clockwise or anti clockwise, horizontal or vertical flipping, horizontal or vertical shifting of image samples.

Adam optimizer has been used for optimization purposes with a learning rate of about 0.0001 for Convoliution Neural Network (CNN) model.The model was first trained for 22 epochs with a batch size of 15. In this model we are using channel last architecture model as a default architecture but a backend switch is also there which can support channel last architecture. Then the first step of CNN is created as Convolution followed by Recified Linear Units (ReLUs) followed by Pooling. Convolution layer of this model has 32-filters with a 3 x 3 kernel and Recified Linear Units (ReLUs) activation.Then the next step is applying batch normalization, max pooling, and 25% (0.25) dropout (regularization technique to overcome the problem in neural networks). When dropout is 0.25, the accuracy increses to 90%. Fig. 6 shows the overall summary of the CNN model implemented with various parameters which is used to predict the size of the model. Fig 7 shows the model history of the complete system. All the experiment analyses were performed on Intel Core i3-7010U CPU.

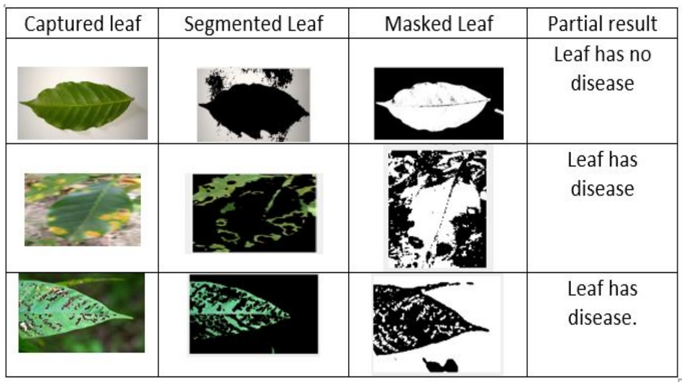
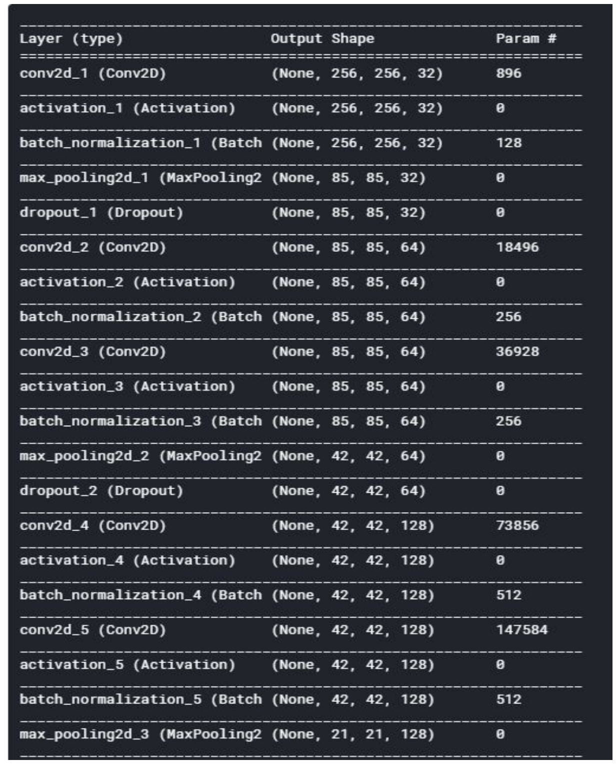


Fig. 5: Image Samples Segmented

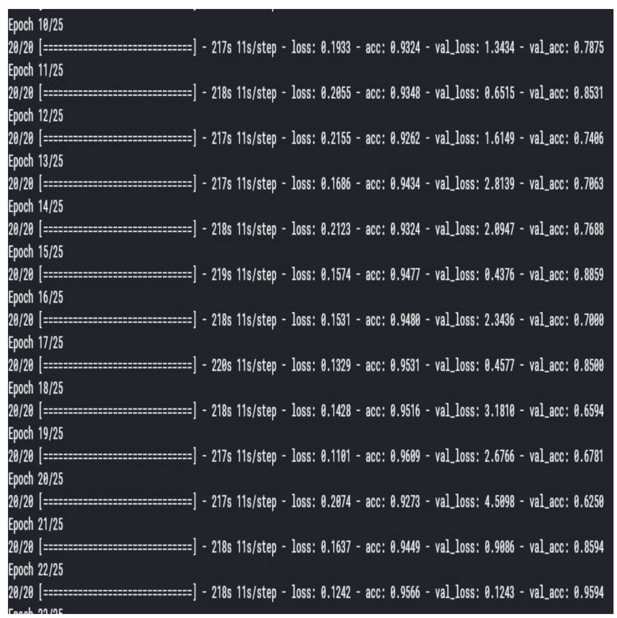
V. Results and Graph

The whole dataset samples is divided into 80% for the training, 20% for validation and testing and all the datasets are collected from two different conditions. One in same dataset samples used for training and testing purposes. Other is that on field condition where images are taken from real time scenario from land. Since the background and lighting of the image samples obtained from real scenario may totally different from the already obtained samples hence there is a probability that the model can show low accuracy when compared with the accuracy of the already trained samples. So to reduce this issue, we can add a mixed variety of image samples during the



Layer (type)	Output Shape	Param #
conv2d_1 (Conv2D)	(None, 256, 256, 32)	896
activation_1 (Activation)	(None, 256, 256, 32)	0
batch_normalization_1 (Batch	(None, 256, 256, 32)	128
max_pooling2d_1 (MaxPooling2	(None, 85, 85, 32)	0
dropout_1 (Dropout)	(None, 85, 85, 32)	0
conv2d_2 (Conv2D)	(None, 85, 85, 64)	18496
activation_2 (Activation)	(None, 85, 85, 64)	0
batch_normalization_2 (Batch	(None, 85, 85, 64)	256
conv2d_3 (Conv2D)	(None, 85, 85, 64)	36928
activation_3 (Activation)	(None, 85, 85, 64)	0
batch_normalization_3 (Batch	(None, 85, 85, 64)	256
max_pooling2d_2 (MaxPooling2	(None, 42, 42, 64)	0
dropout_2 (Dropout)	(None, 42, 42, 64)	0
conv2d_4 (Conv2D)	(None, 42, 42, 128)	73856
activation_4 (Activation)	(None, 42, 42, 128)	0
batch_normalization_4 (Batch	(None, 42, 42, 128)	512
conv2d_5 (Conv2D)	(None, 42, 42, 128)	147584
activation_5 (Activation)	(None, 42, 42, 128)	0
batch_normalization_5 (Batch	(None, 42, 42, 128)	512
max_pooling2d_3 (MaxPooling2	(None, 21, 21, 128)	0

Fig. 6: No of parameters obtained for CNN model



```
Epoch 10/25
20/20 [==============================] - 217s 11s/step - loss: 0.1933 - acc: 0.9324 - val_loss: 1.3434 - val_acc: 0.7875
Epoch 11/25
20/20 [==============================] - 218s 11s/step - loss: 0.2055 - acc: 0.9348 - val_loss: 0.6515 - val_acc: 0.8531
Epoch 12/25
20/20 [==============================] - 217s 11s/step - loss: 0.2155 - acc: 0.9262 - val_loss: 1.6149 - val_acc: 0.7406
Epoch 13/25
20/20 [==============================] - 217s 11s/step - loss: 0.1686 - acc: 0.9434 - val_loss: 2.8139 - val_acc: 0.7063
Epoch 14/25
20/20 [==============================] - 218s 11s/step - loss: 0.2123 - acc: 0.9324 - val_loss: 2.0947 - val_acc: 0.7688
Epoch 15/25
20/20 [==============================] - 219s 11s/step - loss: 0.1574 - acc: 0.9477 - val_loss: 0.4376 - val_acc: 0.8859
Epoch 16/25
20/20 [==============================] - 218s 11s/step - loss: 0.1531 - acc: 0.9480 - val_loss: 2.3436 - val_acc: 0.7000
Epoch 17/25
20/20 [==============================] - 220s 11s/step - loss: 0.1329 - acc: 0.9531 - val_loss: 0.4577 - val_acc: 0.8500
Epoch 18/25
20/20 [==============================] - 218s 11s/step - loss: 0.1428 - acc: 0.9516 - val_loss: 3.1810 - val_acc: 0.6594
Epoch 19/25
20/20 [==============================] - 217s 11s/step - loss: 0.1101 - acc: 0.9609 - val_loss: 2.6766 - val_acc: 0.6781
Epoch 20/25
20/20 [==============================] - 217s 11s/step - loss: 0.2074 - acc: 0.9273 - val_loss: 4.5098 - val_acc: 0.6250
Epoch 21/25
20/20 [==============================] - 218s 11s/step - loss: 0.1637 - acc: 0.9449 - val_loss: 0.9086 - val_acc: 0.8594
Epoch 22/25
20/20 [==============================] - 218s 11s/step - loss: 0.1242 - acc: 0.9566 - val_loss: 0.1243 - val_acc: 0.9594
```

Fig. 7: Full History of the Model

training phase which is known as heterogeneity. From Table No. II, we can say that the validation accuracy gives the best result of 95.94% accuracy when number of epochs is 22. We have also recorded the time taken for running the algorithm in different epochs which is tabulated in Table No. III

TABLE II: Accuarcy comparison with epochs

No of Epochs	Training Accuracy (%)	Validation Accuracy (%)	Testing Accuracy (%)
6	89.73	70.63	70.62223
10	93.24	78.75	78.74578
17	95.31	85.00	84.7567
21	94.49	85.94	85.9376
22	95.66	95.94	95.9381

TABLE III: Time taken by the system

No of Epochs	Training with images	Time taken (mins)
6	1600	14.8
10	1600	30.6
17	1600	55.6
21	1600	70.7
22	1600	95.6

Below graph shows how training and validation accuracy increases and training and validation loss decreases with increase in no of epochs.(x axis denotes no of epochs and y axis denotes accuracy in Fig. 8 and loss in Fig. 9). Finally calculated the accuracy of my model using the validation data which is approximately 95.9381%

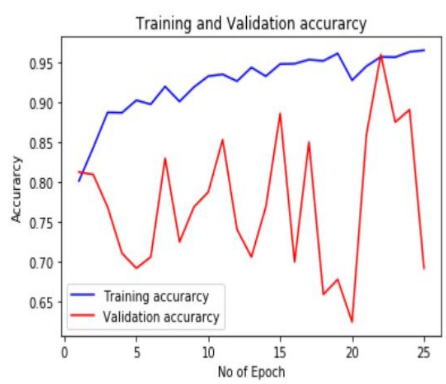


Fig. 8: Plot of accuracy vs no. of epochs

A. Evaluation Matrix

The classification of Tea leaf diseased plant performance of CNN technique which have been analyzed for near about 1300 input leaf images. The performance evaluated for the neural network techniques which have been used in this thesis from the confusion matrix shown below of their respective classifier.

Table IV shows that the overall confusion matrix of model of different disease classification for the 3 classes. It clearly depicts that class 0 which is healthy tea leaf is 15% time mis classified as class 1 which is rust of tea and by 5% as class 2 which is black rot of tea. Class 1(Rust of Tea) is 10%

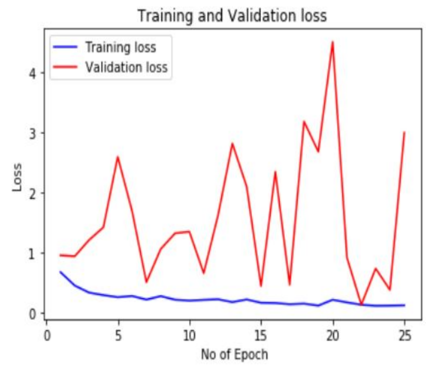


Fig. 9: Plot of loss vs no. of epochs

TABLE IV: Classification Matrix

		Predicted Class		
		Class 0	**Class 1**	**Class 2**
Actual Class	**Class 0**	0.80	0.15	0.05
	Class 1	0.10	0.76	0.14
	Class 2	0.03	0.11	0.86

mis classified as class 0(Healthy Leaf) and by 14% as class 2(Black Rot). Class 2(Black rot) 3% mis classified as class 0(Healthy Leaf) and by 11% as class 1(Rust of Tea).

We have then shown the classification report, which has precision, recall, and F1-score for all the 3 classes. This is shown in Fig IV. Precision is mainly the ability of the classifier to predict the real positive images as positive and recall shows the amount of positive images that are classified correctly as positive. Precision and recall calculated by the following formula:

$$Precision = \frac{TruePositive}{TruePositive + FalsePostive} \tag{1}$$

$$Recall = \frac{TruePositive}{TruePositive + Negative} \tag{2}$$

TABLE V: Classification Matrix

	Precision	Recall	F1- Score	Support
Class 0	0.50	0.50	0.50	10
Class 1	0.60	0.75	0.67	8
Class 2	0.40	0.29	0.33	7
Micro Average	0.52	0.52	0.52	25
Macro Average	0.50	0.51	0.50	25
Weighted Average	0.50	0.52	0.51	25

The classification matrix is shown in Table V. The F1 score (also F-score or F-measure) is a measure of a test's accuracy. It is calculated from the precision and recall of the test, where the precision is the number of correctly identified positive results divided by the number of all positive results, including those not identified correctly, and the recall is the number of correctly identified positive results divided by the number of all samples that should have been identified as positive. Mathematically, its value is:

$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (3)$$

VI. Conclusion

Generating and training of CNN model from onset is a very monotonous process when compared to existing deep learning methods to gain maximum accuracy. Our result shows that model when trained with 22 number of epoch cycle gives the highest accuracy using tensor flow. In this paper only 2 types of tea leaf disease images including one healthy leaf disease have been classified using 3 layer CNN model, but in future we can increase the number of classes and modify the architecture accordingly to gain better accuracy. We can also use an efficient model than this existing architecture on real time data for mobile based IoT application.

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